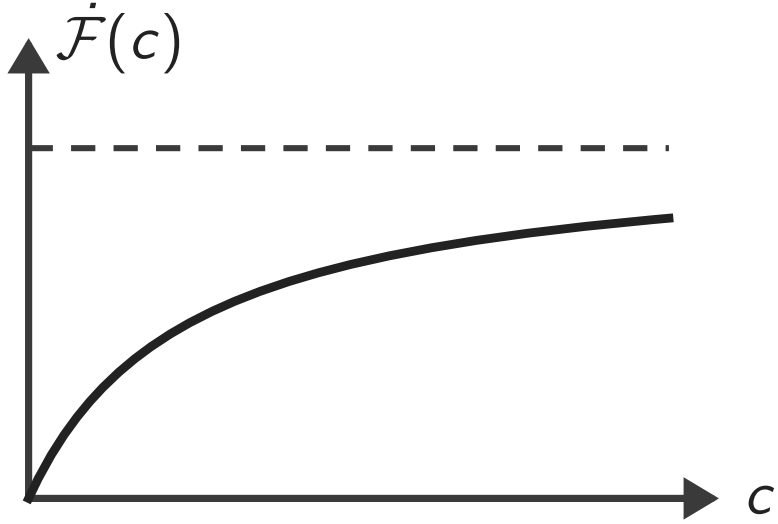
$$V = \frac{1}{2}\left(\frac{\delta c}{c}\right)^2 \underbrace{\dot{\mathcal{F}}(c)}$$

Fisher information rate



$$\dot{\mathcal{F}}(c) = -\lim_{t\to\infty} \frac{1}{t}\left\langle \frac{\partial^2 \ln P(\mathcal{O}(t)\,|\,c)}{(\partial \ln c)^2} \right\rangle_c$$

$$= \frac{1}{\tau_{\text{burst}}}$$

$c\,k_b$: diffusion-limited arrival of molecules

$$k_b = 4aD$$